$$\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{\left(\sin^2 x + \cos^2 x\right)^2 - 2\sin^2 x \cos^2 x} dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin 2x}{1 - \frac{1}{2}\sin^2 2x} dx$$

$$u \coloneqq \cos 2x$$

$$du = -2\sin 2x dx$$

$$\cos^2 2x = u^2$$

$$\sin^2 2x = 1 - u^2$$

$$\frac{1}{2}\frac{\int_{-1}^{1}}{1 - \frac{1}{2}(1 - u^2)} du$$

$$\frac{1}{2}\frac{\int_{-1}^{1}}{1 - \frac{1}{2} - \frac{1}{2}u^2} du$$

$$\frac{1}{2}\frac{\int_{-1}^{1}}{\frac{1}{2} - \frac{1}{2}u^2} du$$

$$\frac{\int_{-1}^{1}}{1 - u^2} du$$

$$\arctan u \,\Big|_{-1}^{1}$$

$$\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)$$

$$\frac{\pi}{2}$$